Customer Churn and Revenue Retention Analysis

Executive summary

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Purpose and decisions requested

Customers are leaving. Leadership asked why, how much recurring revenue is going with them, and whether anything we already record could help us keep more of them. This summary reports what our customer data shows and what it cannot show. It describes patterns and does not predict who will leave.

Three decisions are requested:

- **How competitor-offer churn is counted.** Counting it as something the company could have influenced puts the influenceable revenue at about $1.48 million a year. Not counting it puts the figure at about $923,000 a year.
- **Who investigates the gaps customers named.** Customers who left for a competitor cited speeds, data allowances, devices and offers. Pricing and product are best placed to check what is behind this.
- **How the proposed tests are judged.** One follow-up period and a comparison group need to be agreed before any of the tests described near the end are run.

How to read these figures

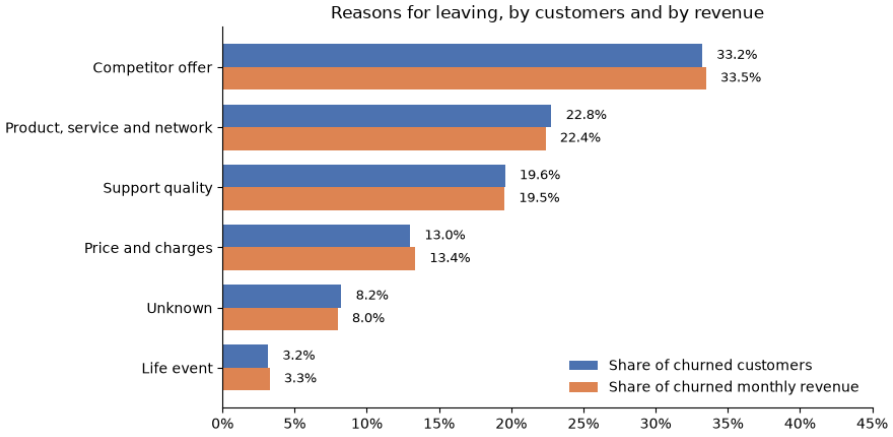
Three points apply to everything that follows.

- **Annual figures are estimates.** Each is the monthly charges of churned customers multiplied by 12. It is not recognized revenue and not an amount the company could be sure to recover.
- **Findings are associations.** The data is a single snapshot, so patterns show what goes together and do not show cause.
- **Reasons are self-reported.** The reasons customers gave for leaving are their own account and were not checked against other records.

The scale of the problem

26.5% of customers (1,869 of 7,043) have churned. They hold 30.5% of all monthly charges, which is more than their share of customers because churned customers pay more on average. The monthly charges of churned customers, multiplied by 12, come to an estimated $1.67 million a year in lost recurring revenue.

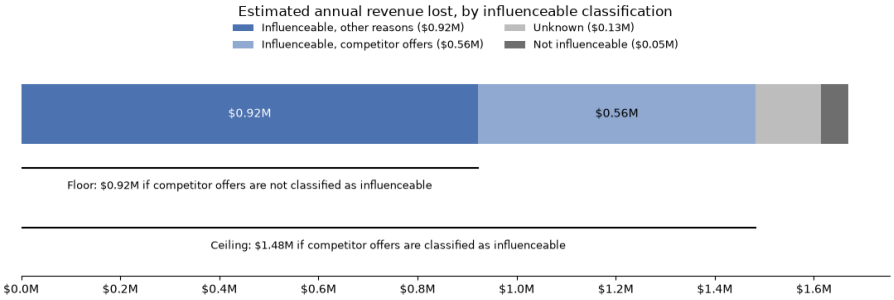
Why customers say they left



Competitor offers are the largest group: 621 customers, 33.2% of churners and about $559,000 a year. Product, service and network issues follow at 22.8% and support quality at 19.6%. Price and charges account for 13.0%. Another 154 customers (8.2%) gave no usable reason, and only 59 (3.2%) left for life events.

The customer bars and the revenue bars are almost the same height in every group, so weighting by revenue does not change the ranking. The reasons are self-reported. They describe what customers said and are not proven causes. The price share in particular may be understated if some competitor-offer answers are really price decisions.

How much of the loss the company could have influenced



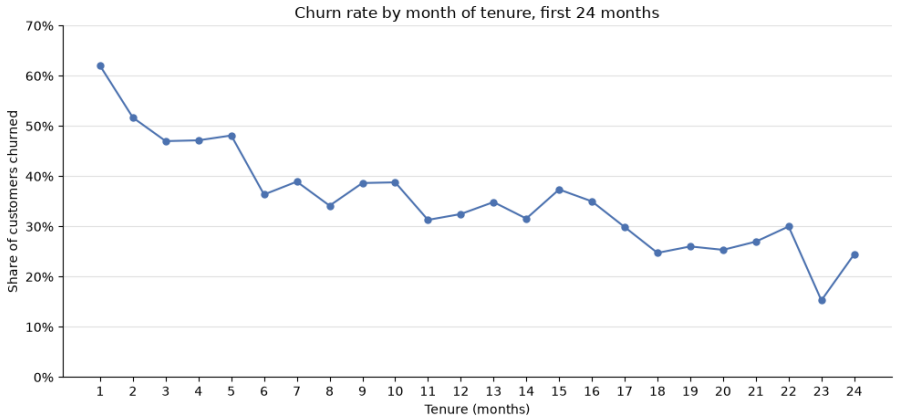
Each reason was classified as influenceable (the company had a lever it could have used), not influenceable (life events such as moving or death) or unknown (no usable reason). Competitor offers are the debatable group. The competitor's action is outside the company's control, but speeds, data allowances, device offers and pricing are company decisions.

With competitor offers counted as influenceable, 1,656 customers (88.6%) and about $1.48 million a year were influenceable. Without them, 1,035 customers (55.4%) and about $923,000 a year remain. The gap of 621 customers and about $559,000 a year is the competitor-offer group itself. The $923,000 is influenceable under either classification, so it is the floor, and $1.48 million is the ceiling.

The high share under the first classification reflects the definition as much as the data, because four of the six reason groups are classed as influenceable by definition. The informative findings

are the small share lost to life events and the size of the range. Influenceable does not mean the customer would have stayed, and it does not mean responding would have been worth the cost.

When customers leave

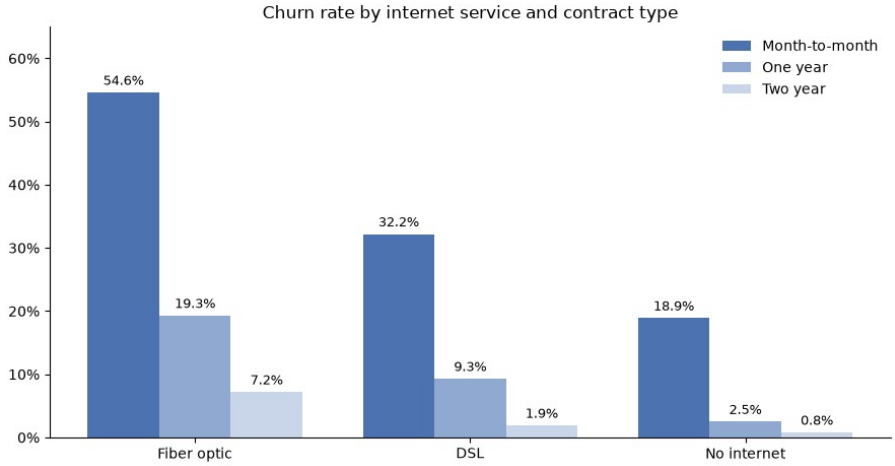


Churn is highest in the first month of tenure. Of 613 customers recorded at one month, 380 (62.0%) have churned, and these customers alone are 20.3% of all churners. The rate falls to 51.7% in month 2, holds at about 47% to 48% in months 3 to 5, and steps down to 36.4% at month 6. The chart covers the first 24 months, which hold 71.2% of all churners.

By band, churn is 47.4% in the first year, then 28.7%, 20.4% and 9.5% in later bands. The later bands are mostly customers who stayed, because those who were going to leave early are already gone. More than half of all churners (1,037 of 1,869, 55.5%) left within the first year, about $827,000 a year or 49.6% of the estimated total.

The reasons customers gave do not explain the early spike. Support quality is cited by 19.9% of churners in months 1 to 5 and 19.4% from month 6 onward, and no reason group shows a statistically significant difference between early and later churners, even after adjusting for the six comparisons. This does not prove the two groups give identical reasons, but it means the stated reasons give no sign that early losses are an onboarding or support problem in particular. The cause of early churn is not known from this data.

Who leaves: contract type and internet service

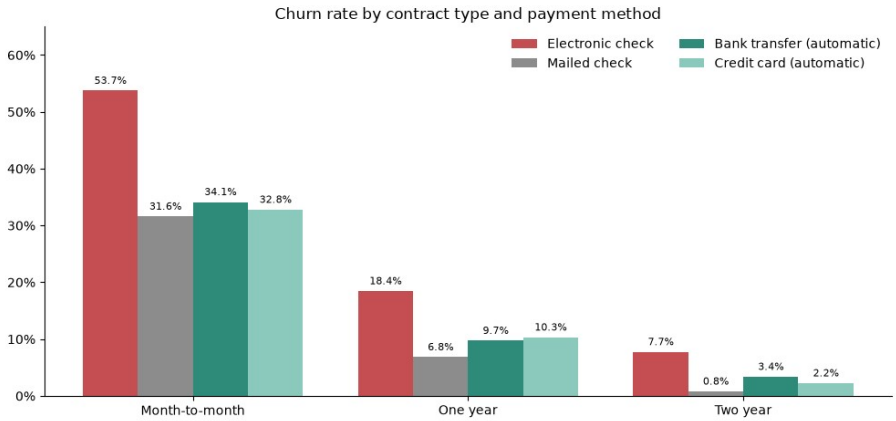


Churn is concentrated in two overlapping groups. Month-to-month customers churn at 42.7%, against 11.3% on one-year and 2.8% on two-year contracts. They are 55.0% of customers but 88.6% of churners. Fiber customers churn at 41.9%, against 19.0% for DSL and 7.4% for customers without internet. They are 69.4% of churners and 82.2% of churned revenue, because they also pay more ($91.50 a month on average, against $58.10 for DSL).

Each pattern holds inside the other, as the chart shows: within every internet service, month-to-month customers churn far more, and within every contract type, fiber churns more than DSL. The highest-risk group is fiber customers on month-to-month contracts: 2,128 customers (30.2% of the base), 54.6% of whom churned, producing 62.2% of all churners.

In the first five months, 96.9% of customers are on month-to-month contracts, and 74.8% of the fiber customers in that window had churned, against 50.3% for DSL and 24.4% for customers without internet. These are associations in a single snapshot. Customers who plan to stay may choose longer contracts, so the data shows where churn concentrates and not what causes it.

Payment method

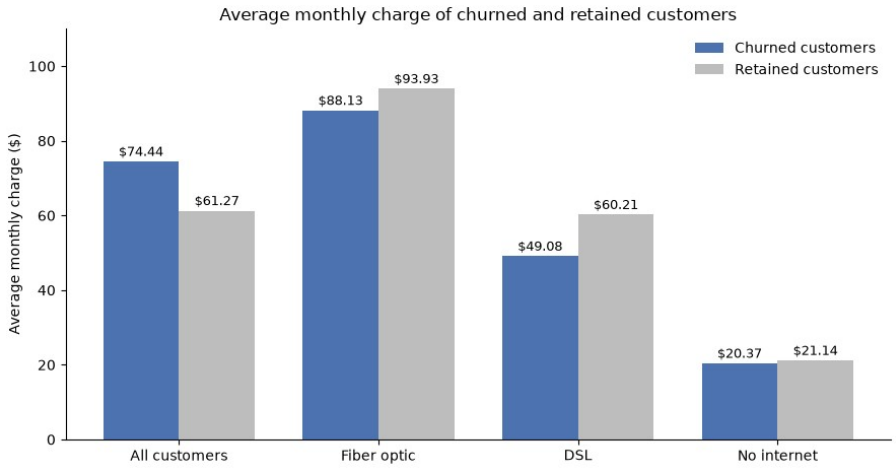


Electronic check customers churn at 45.3% (1,071 of 2,365), against 19.1% for mailed check, 16.7% for bank transfer and 15.2% for credit card. They account for 57.3% of churners and about $1.01 million a year. They are also more often on month-to-month contracts (78.2%), so the comparison was repeated within each contract type. The gap holds in all three: on month-to-

month contracts, 53.7% against 31.6% to 34.1%; on one-year contracts, 18.4% against 6.8% to 10.3%; and on two-year contracts, 7.7% against 0.8% to 3.4%.

Among month-to-month customers the gap also holds within fiber and within DSL and in every tenure band. Customers without internet are the exception, but those groups are small. On month-to-month contracts, customers paying by an automatic method do not show lower churn than customers paying by mailed check. The data holds no record of failed payments, billing disputes or when a customer chose or changed their payment method, so it cannot show why electronic check customers leave.

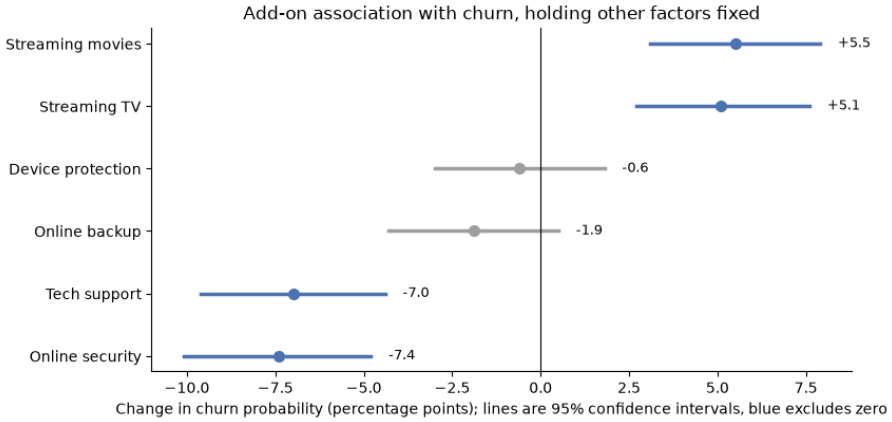
Monthly charges



Across all customers, churned customers pay more: $74.44 a month against $61.27 for retained customers. Within every internet service the direction reverses. Fiber churners pay $88.13 against $93.93 for those who stayed, and DSL churners $49.08 against $60.21. For customers without internet the gap is 77 cents, which is statistically detectable but too small to matter commercially. The overall average is higher for churners because most of them (69.4%) are on fiber, the most expensive service.

Within fiber and within DSL, observed churn is lower in the higher charge quarters (from 55.7% to 26.3% on fiber and from 33.4% to 6.8% on DSL). Once contract type is also held fixed, the differences between churned and retained customers are small and inconsistent in direction, so the monthly bill shows no consistent link to churn in this data. The data holds no competitor prices or discount history, so it cannot show whether prices were too high compared with alternatives.

Add-on services

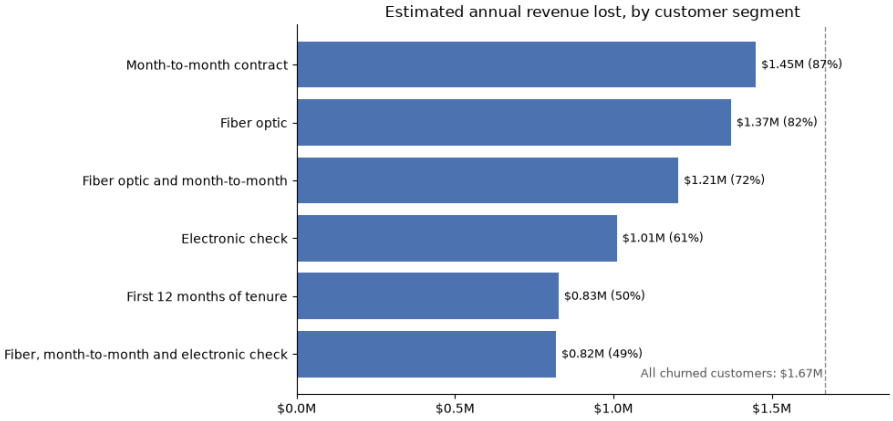


Churn falls steadily with the number of add-ons, from 52.2% for customers with none to 5.3% for customers with all six, but this is mostly a tenure effect. Average tenure rises from 10 months to 65 across the same steps, and once tenure is held fixed most of the pattern disappears.

The chart shows each add-on separately, from a model that accounts for contract type, internet service and tenure across the 5,517 customers with internet service. Online security and tech support are associated with about 7 points lower churn (7.4 and 7.0). Streaming TV and streaming movies are associated with about 5 points higher churn (5.1 and 5.5). Online backup and device protection show no clear difference. The four clear results stay clear after adjusting for the six comparisons, and they barely move when tenure is entered as bands. Online backup does move (to 2.6 points lower, barely distinguishable from zero), so its result is small and should be read cautiously.

Looking at tech support alone, month-to-month internet customers with it churn 13.5 to 20.4 points less in every tenure band, with fiber and DSL customers combined. These are associations. Customers who take these services may already be less likely to leave, so the data points to where a retention test could be run and does not show that offering the services would reduce churn.

Where the lost revenue is concentrated



Month-to-month customers account for $1.45 million of the estimated $1.67 million (86.9%), and fiber customers for $1.37 million (82.2%). Electronic check customers account for $1.01

million (60.6%) and customers in their first 12 months for $0.83 million (49.6%). One segment stands out: fiber customers on month-to-month contracts paying by electronic check are 18.6% of customers, churn at 60.4%, and hold 42.2% of churners and 49.1% of the lost revenue.

The segments overlap, so their figures are not additive and should not be summed. They describe where revenue was lost and are not separate amounts that could each be recovered.

What the data could not answer

- **Why customers left.** Reasons are self-reported, and whether a stated reason is the real cause cannot be checked.
- **Competitor pricing, discounts and promotions.** The data has no competitor prices or offers beyond the reason customers gave, and no history of discounts or price changes.
- **Retention efforts.** There are no records of offers made to customers or how they responded, so the effect of any action cannot be measured.
- **Payment problems.** The higher churn among electronic check customers cannot be explained without records of failed payments, billing disputes, or when customers chose or changed their payment method.
- **Service quality and location.** There is no data on network quality, outages or support contacts, and location is too sparse to use (a median of 4 customers per city).

Answering these would need billing and payment event data, retention campaign records, service quality data and customer-level location. Gender and senior citizen status were left out because retention decisions based on protected characteristics raise fairness concerns.

Recommended next steps

The findings point to five areas the company could test. None of them estimates how much churn a response would prevent. Only a controlled test, with a comparison group that does not receive the change, could show that, so these are hypotheses to check and not established interventions. Before any test, one follow-up period should be fixed and used for every success measure so the results can be compared.

Recommendation on the first decision. Until pricing and product confirm which competitor-offer losses were addressable, plan on the $923,000 floor and report the competitor-offer revenue (about $559,000 a year) separately. This is a planning judgment and not a finding of the analysis.

Area	What the company could test	Success measure	What could undermine it
Competitor-offer churn	Investigate the product, pricing and service gaps these customers named, then test targeted retention offers against a comparison group.	Share of customers retained over a defined follow-up period, offered group against comparison group, and cost per customer retained.	For some customers the cost of a response may exceed the revenue kept.

Area	What the company could test	Success measure	What could undermine it
First year of the relationship	Test an improved first-year experience for new customers, starting with fiber, against a comparable group that does not receive it.	First-year churn rate of the tested group against the comparison group.	The reasons customers gave show no detectable difference by tenure, so the cause of early churn is not known and the change may not help.
Fiber customers on month-to-month contracts	Look into the fiber service experience and test contract options for these customers.	Churn rate over a fixed period for customers offered the option against those not offered it.	Customers who choose longer contracts may already be less likely to leave.
Electronic check customers	Review payment records for failed payments and billing disputes, which this data does not hold, then test a change to the payment experience.	Primary: churn over a defined period, offered against not offered. Secondary: payment failure rate and billing disputes. Guardrail: cost per customer retained.	The cause may not be the payment method, and the records may show nothing.
Tech support and online security	Offer these services to a sample of customers and compare with a group not offered them.	Churn rate and cost per customer retained.	Customers who take these services may already be less likely to leave, and the cost may exceed the revenue kept.

Notes on the figures and the data

The data is IBM's public Telco Customer Churn sample: 7,043 customers in a single snapshot. The sample is synthetic, so the results demonstrate method and are not findings about a real company. Churn means the customer is recorded as having left during the quarter (Q3) described in the dataset documentation, so the rates describe that quarter.

Group differences were checked with chi-square and Mann-Whitney tests, with a Holm adjustment where related tests were run. Add-on effects come from a logistic regression with a robustness check. The calculations behind every figure are in the project notebook at github.com/FatihaOkesola/telco-churn-revenue-analysis.